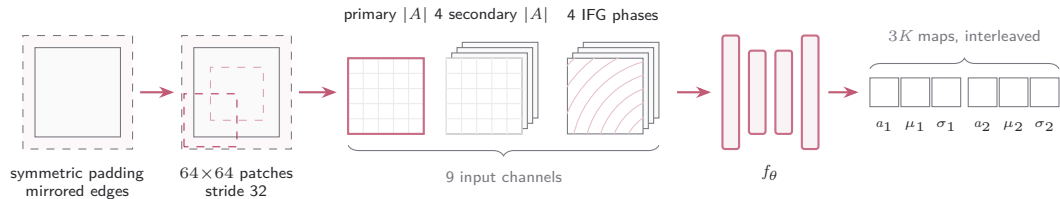


DLR-TomoSAR — Input & output representation

nine channels · two regimes · per-channel normalization

July 2026

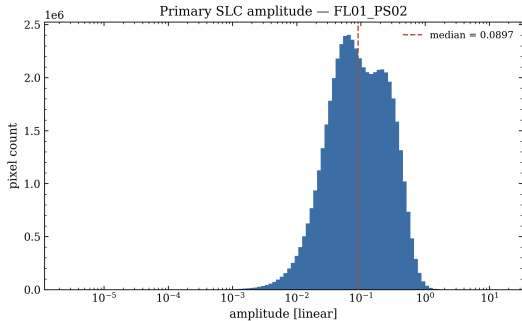
Input and output representation



- Scenes are symmetrically padded (edge-mirrored) so the 64×64 , stride-32 patch grid tiles them exactly; sliding-window stitching undoes the overlap at inference.
- The $3K$ output channels are interleaved per scatterer — $a_1, \mu_1, \sigma_1, a_2, \mu_2, \sigma_2, \dots$ — not grouped by parameter type.
- Outputs live in normalized space; denormalised and clamped to physical bounds before any curve is reconstructed.
- Current configuration $K=2$ (6 maps); the $K=5$ variant (15 maps) in development.

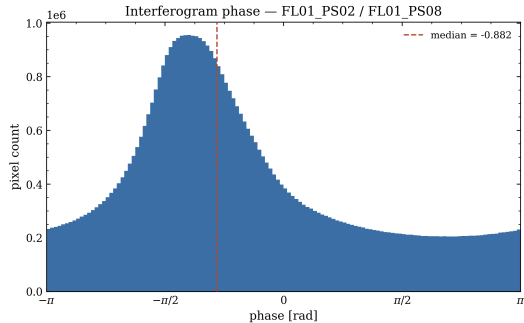
Two input channels, two regimes

primary amplitude



Log-scaled axis: the bulk sits near 10^{-1} but the tail runs past 10^0 — orders of magnitude in one channel. **Heavy-tailed** → robust log1p scaling.

interferogram phase



Confined to $[-\pi, \pi]$ by construction, a single broad mode — no tail to tame. **Bounded by physics** → fixed $\div \pi$.

Real F-SAR channels (primary PS02, IFG PS02/PS08). The nine inputs split cleanly into these two shapes — and the normalization strategy (next) follows the split.

Per-channel normalization — old → new, channel by channel

z-score — standard score · IQR — interquartile range

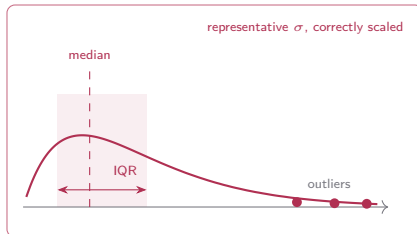
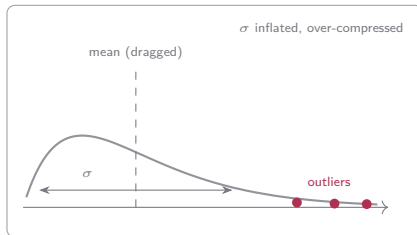
channel	old	new	why
pass / mag	z-score (log1p)	→ robust IQR (log1p)	tail sets μ , σ
ifg / phase	min-max ($p_{99.9}$)	→ fixed $\div \pi$	bounded by physics
out / a	z-score	→ robust IQR (log1p)	heavy-tailed target
out / σ	z-score	→ robust IQR (log1p)	heavy-tailed target
out / μ	z-score	= z-score (<i>kept</i>)	bounded to sampling range

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

old — moments fit the tail. Rare bright pixels drag μ and inflate σ ; the bulk compresses into a narrow band.

$$\tilde{x} = \frac{\log(1+x) - \text{med}}{Q_3 - Q_1}$$

new — quantiles fit the bulk. The median and IQR ignore the tail; the bulk keeps its spread.



Normalization ablation — every rung improves, the output channels most

metric	base	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out σ IQR log1p
curve R^2 $\uparrow$	0.51 $\pm$.04	0.53 $\pm$.01	0.54 $\pm$.01	0.58 $\pm$.03	0.57 $\pm$.02
curve MAE $\downarrow$	0.0572 $\pm$.003	0.0544 $\pm$.001	0.0531 $\pm$.001	0.0504 $\pm$.002	0.0501 $\pm$.001
PSNR [dB] $\uparrow$	49.8 $\pm$.3	49.9 $\pm$.1	50.1 $\pm$.1	50.5 $\pm$.3	50.4 $\pm$.2
per-pixel R^2 $\uparrow$	0.070 $\pm$.015	0.084 $\pm$.005	0.091 $\pm$.004	0.323 $\pm$.031	0.359 $\pm$.050
profile cos med $\uparrow$	0.000 $\pm$.000	0.000 $\pm$.000	0.000 $\pm$.000	0.847 $\pm$.027	0.851 $\pm$.020
SSIM elevation $\uparrow$	0.953 $\pm$.002	0.955 $\pm$.001	0.956 $\pm$.001	0.958 $\pm$.002	0.958 $\pm$.001
SSIM range $\uparrow$	0.905 $\pm$.003	0.907 $\pm$.001	0.909 $\pm$.001	0.923 $\pm$.001	0.926 $\pm$.003
SSIM azimuth $\uparrow$	0.933 $\pm$.003	0.935 $\pm$.001	0.937 $\pm$.001	0.944 $\pm$.002	0.946 $\pm$.001
matched F1 $\uparrow$	0.467 $\pm$.009	0.472 $\pm$.004	0.476 $\pm$.003	0.688 $\pm$.025	0.727 $\pm$.040
matched recall $\uparrow$	0.351 $\pm$.006	0.354 $\pm$.004	0.356 $\pm$.003	0.604 $\pm$.036	0.656 $\pm$.056
count exact $\uparrow$	0.246 $\pm$.005	0.246 $\pm$.005	0.252 $\pm$.002	0.547 $\pm$.050	0.611 $\pm$.066
peak err mean $\downarrow$	10.7 $\pm$.1	10.7 $\pm$.1	10.7 $\pm$.1	5.2 $\pm$.8	4.5 $\pm$.9
median $\downarrow$	13.7 $\pm$.4	13.8 $\pm$.4	13.7 $\pm$.4	1.2 $\pm$.6	0.9 $\pm$.4
p95 $\downarrow$	16.8 $\pm$.0	16.8 $\pm$.0	16.8 $\pm$.0	15.8 $\pm$.6	15.4 $\pm$.7
matched μ MAE $\downarrow$	3.44 $\pm$.09	3.37 $\pm$.05	3.30 $\pm$.02	2.47 $\pm$.10	2.30 $\pm$.12
matched σ MAE $\downarrow$	1.39 $\pm$.03	1.36 $\pm$.02	1.34 $\pm$.01	0.96 $\pm$.04	0.86 $\pm$.05
matched a MAE $\downarrow$	0.503 $\pm$.014	0.503 $\pm$.008	0.503 $\pm$.013	0.383 $\pm$.020	0.364 $\pm$.034

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas per rung. **bold** = best in row, marked only where best and worst differ by more than 5%; **red heads** = the output rungs; **green** = the metrics those rungs unlock. Base = z-score outputs, zscore-log1p pass, minmax-p99 ifg (out μ stays z-score). μ , σ , peak in elevation units; a in linear amplitude. Peak median/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

Normalization ablation — the gains concentrate on single-scatterer pixels

metric	base	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out σ IQR log1p
matched precision $\uparrow$	0.699 $\pm$.015	0.709 $\pm$.005	0.720 $\pm$.005	0.800 $\pm$.008	0.816 $\pm$.015
$k=1$	0.602 $\pm$.015	0.604 $\pm$.014	0.624 $\pm$.005	0.827 $\pm$.020	0.848 $\pm$.019
$k=2$	0.752 $\pm$.016	0.765 $\pm$.007	0.771 $\pm$.005	0.767 $\pm$.017	0.776 $\pm$.013
matched recall $\uparrow$	0.351 $\pm$.006	0.354 $\pm$.004	0.356 $\pm$.003	0.604 $\pm$.036	0.656 $\pm$.056
$k=1$	0.182 $\pm$.006	0.180 $\pm$.006	0.182 $\pm$.004	0.562 $\pm$.068	0.647 $\pm$.088
$k=2$	0.591 $\pm$.013	0.602 $\pm$.007	0.604 $\pm$.006	0.663 $\pm$.014	0.670 $\pm$.016
count acc pred $k=1$ $\uparrow$	0.763 $\pm$.021	0.781 $\pm$.015	0.796 $\pm$.013	0.942 $\pm$.009	0.952 $\pm$.009
count acc pred $k=2$ $\uparrow$	0.694 $\pm$.008	0.698 $\pm$.006	0.711 $\pm$.007	0.717 $\pm$.012	0.720 $\pm$.013
matched μ MAE $\downarrow$	3.44 $\pm$.09	3.37 $\pm$.05	3.30 $\pm$.02	2.47 $\pm$.10	2.30 $\pm$.12
$k=1$	1.25 $\pm$.06	1.15 $\pm$.05	1.10 $\pm$.03	0.63 $\pm$.05	0.63 $\pm$.06
$k=2$	4.18 $\pm$.11	4.10 $\pm$.04	4.03 $\pm$.03	4.17 $\pm$.10	4.07 $\pm$.05
matched σ MAE $\downarrow$	1.39 $\pm$.03	1.36 $\pm$.02	1.34 $\pm$.01	0.96 $\pm$.04	0.86 $\pm$.05
$k=1$	0.61 $\pm$.02	0.59 $\pm$.02	0.59 $\pm$.01	0.30 $\pm$.02	0.25 $\pm$.02
$k=2$	1.65 $\pm$.03	1.61 $\pm$.01	1.58 $\pm$.01	1.58 $\pm$.04	1.51 $\pm$.01
matched a MAE $\downarrow$	0.503 $\pm$.014	0.503 $\pm$.008	0.503 $\pm$.013	0.383 $\pm$.020	0.364 $\pm$.034
$k=1$	0.356 $\pm$.020	0.359 $\pm$.016	0.351 $\pm$.018	0.169 $\pm$.018	0.155 $\pm$.029
$k=2$	0.552 $\pm$.012	0.550 $\pm$.005	0.554 $\pm$.011	0.581 $\pm$.021	0.586 $\pm$.035

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%; **red heads** = the output rungs; **green** = the $k=1$ collapse they buy, **red** = the one drift up. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

Normalization ablation — slot 0 calibrates, slot 1 trades leakage against amplitude

prediction mean	GT	base	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out σ IQR log1p
<i>slot 0 — strongest scatterer; GT-active on every pixel</i>						
active frac	1.00	0.344 $\pm$.004	0.342 $\pm$.004	0.342 $\pm$.003	0.644 $\pm$.051	0.706 $\pm$.066
a — active	0.77	1.87 $\pm$.03	1.88 $\pm$.03	1.87 $\pm$.01	0.98 $\pm$.08	0.89 $\pm$.07
μ — active	-4.89	-3.73 $\pm$.10	-3.58 $\pm$.07	-3.59 $\pm$.02	-4.00 $\pm$.09	-4.03 $\pm$.15
σ — active	1.76	2.36 $\pm$.05	2.31 $\pm$.05	2.36 $\pm$.03	1.73 $\pm$.04	1.52 $\pm$.08
<i>slot 1 — second scatterer; GT-active on 26% of pixels</i>						
active frac	0.260	0.289 $\pm$.005	0.288 $\pm$.004	0.281 $\pm$.003	0.308 $\pm$.007	0.306 $\pm$.008
a — active	1.43	1.33 $\pm$.04	1.33 $\pm$.03	1.34 $\pm$.03	1.04 $\pm$.04	1.06 $\pm$.05
a — inact.	0.00	0.74 $\pm$.03	0.70 $\pm$.03	0.68 $\pm$.01	0.44 $\pm$.02	0.45 $\pm$.04
μ — active	11.34	10.39 $\pm$.39	10.20 $\pm$.38	10.56 $\pm$.32	10.23 $\pm$.48	9.97 $\pm$.40
μ — inact.	—	3.66 $\pm$.31	3.65 $\pm$.26	3.98 $\pm$.24	3.73 $\pm$.72	3.39 $\pm$.57
σ — active	3.53	3.59 $\pm$.10	3.52 $\pm$.09	3.58 $\pm$.06	3.41 $\pm$.11	3.02 $\pm$.04
σ — inact.	—	2.36 $\pm$.10	2.27 $\pm$.05	2.21 $\pm$.02	2.06 $\pm$.06	1.90 $\pm$.04
<i>distribution width — std over GT-active pixels</i>						
a — slot 0	1.48	1.43 $\pm$.04	1.44 $\pm$.04	1.42 $\pm$.04	1.24 $\pm$.07	1.17 $\pm$.06
μ — slot 0	3.30	2.27 $\pm$.10	2.31 $\pm$.08	2.38 $\pm$.05	2.07 $\pm$.08	2.03 $\pm$.12
σ — slot 0	2.84	0.95 $\pm$.05	0.95 $\pm$.05	1.07 $\pm$.06	1.30 $\pm$.05	1.01 $\pm$.03
a — slot 1	1.45	0.92 $\pm$.03	0.94 $\pm$.04	0.95 $\pm$.03	0.83 $\pm$.04	0.84 $\pm$.05
μ — slot 1	16.24	7.28 $\pm$.23	7.26 $\pm$.21	7.55 $\pm$.23	8.65 $\pm$.29	8.76 $\pm$.27
σ — slot 1	4.01	1.11 $\pm$.05	1.11 $\pm$.03	1.19 $\pm$.04	1.35 $\pm$.08	1.14 $\pm$.04

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the calibration shifts that matter, **red** = the one overshoot past the GT. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.